

Luca Ballotta

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EDUCATION

PhD degree in Information Engineering

2019 – 2022

University of Padova, Padova, Italy

Defense: March 2023

- **Main research activity** (main collaborators between brackets):
 - Resource allocation in edge computing (L. Carlone, MIT; P. Dini, CTTC)
 - Control of delay systems (M. R. Jovanović, USC)
 - Resilient consensus (G. Como, Polito; J. S. Shamma, UIUC)
- **Thesis:** On fundamental tradeoffs and architecture design in networked control systems
- **Advisor:** prof. Luca Schenato

Master's degree in Automation Engineering

2017 – 2019

University of Padova, Padova, Italy

- **Thesis:** Control and estimation of network dynamical systems with computation latency
- **GPA:** 30/30
- **Honors:** Magna Cum Laude

Bachelor's degree in Information Engineering

2014 – 2017

University of Padova, Padova, Italy

- **Thesis:** A new approach to robust transport over networks
- **GPA:** 28.8/30
- **Honors:** Magna Cum Laude

CURRENT POSITION

Assistant professor

Dec. 2025 – present

Dept. Information Engineering, University of Padova, Italy

PREVIOUS POSITIONS

Research Collaborator

Aug. – Nov. 2025

University of Padova, Italy

- **Research activity:** Reinforcement learning for distributed energy resources

Postdoctoral Researcher

Dec. 2023 – June 2025

Delft University of Technology, Delft, Netherlands

- **Research activity** (main collaborators between brackets):
 - Security of control systems (A. J. Gallo, Polimi)
 - Safety-critical fault detection (M. Mazo and A. Peruffo, TUD)
 - Actuator selection and scheduling (G. Joseph, TUD)
 - Resilient distributed learning (K. H. Johansson and N. Bastianello, KTH)

- Responsible person for schedule of group meetings
- **Supervisor:** prof. Riccardo Ferrari

Research Fellow

Nov. 2022 – Oct. 2023

University of Padova, Padova, Italy

- **Research activity** (main collaborators between brackets):
 - Safe distributed multi-robot control (R. Talak and C. Fan, MIT)
 - Resilient multi-robot consensus (M. Yemini, Bar-Ilan U.; S. Gil, Harvard U.)
 - Vehicular federated learning (M. Rossi, Unipd; G. Piro, Poliba)
 - Control of delay systems (V. Gupta, Purdue U.; J. Arbelaiz, Princeton U.)
- **Supervisor:** prof. Luca Schenato

Visiting Student

Spring 2022

Massachusetts Institute of Technology, Cambridge, USA

- Design of algorithms for multi-robot collaboration (L. Carlone)

Visiting Student

Spring 2020

Massachusetts Institute of Technology, Cambridge, USA

- Algorithms for computation-scheduling co-design (L. Carlone, E. Modiano)

RESEARCH PROJECTS

- **PHYSIA** (Netherlands Enterprise Agency) 2024 – 2025
Postdoc Member
- **TWAIN** (Horizon Europe) 2023 – 2024
Participant
- **Realtime Control of 5G Wireless Networks** (Italian MIUR) 2022 – 2023
Participant
- **HiPeR** (CARIPARO Foundation) 2020 – 2022
Participant

Under review:

- **Intelligent Energy Scheduling in Microgrids** (Italian MAECI-China NSFC) 90k€
PI
- **Beyond Assistance: AI as teammate** (Microsoft Fellowship) 27k€
PI
- **LAVA** (NVIDIA Academic Grant Program) Cloud computing + GPUs
PI
- **HiPerDriVE** (NVIDIA Academic Grant Program) Cloud computing + GPUs
Member

AWARDS & HONORS

PhD Award (finalist)

2024

European Embedded Control Institute (EECI)

- Best PhD thesis in systems & control defended in Europe in 2023

Student Travel Support 2022

IEEE Control Systems Society (IEEE CSS)

- Financial support for attendance at the IEEE Conference on Decision and Control

Young Author Prize 2020

International Federation of Automatic Control (IFAC)

- Best paper at IFAC World Congress whose 1st author is <30 years (170+ nominations)

Roberto Rocca Award (finalist) 2018

Tenaris Dalmine SpA

- Best 1st-year student of four engineering MSc. degrees from the University of Padova

UNIVERSITY TEACHING EXPERIENCE

Responsible instructor Spring 2026

Dept. Information Engineering, University of Padova

BSc. course: Systems & Models

Teaching Assistant Spring 2025

DCSC, Delft University of Technology

MSc. course: Nonlinear Control Systems

Daily supervisor Fall 2024

BSc. AI Minor, Delft University of Technology

BSc. course: Capstone Applied AI Project

Daily supervisor Fall 2024

MSc. AI Minor, Delft University of Technology

MSc. course: Advanced Interdisciplinary AI Project

Responsible instructor Spring 2024

Italian National PhD Program in Autonomous Systems (DAUSY)

PhD course: Introduction to Optimal Linear Quadratic Control

Teaching Assistant AA. YY. 2019-2020, 2020-2021, 2021-2022

Faculty of Engineering, University of Padova

Project: University in Prison

Teaching Assistant Spring 2021

Dept. Information Engineering, University of Padova

BSc. course: Introduction to Automatic Control

Teaching Assistant Fall 2020

Dept. Management Engineering, University of Padova

BSc. course: Calculus 0

Teaching Assistant Fall 2019

Dept. Information Engineering, University of Padova

BSc. course: Calculus I

STUDENT SUPERVISION

PhD students:

- Ruizhe Li (visiting PhD student) Sept. 2026 – Sept. 2027
- Jing Liu (visiting, co-supervised by R. Carli) Sept. 2025 – Aug. 2026

Master's theses: 7

Bachelor's theses: 1

THIRD MISSION

ScegliConNoi

Feb. 2026

Padova Fair, Padova

Information desk of Unipd Department of Information Engineering for high school students

Connect to High Tech at Mechanical Engineering

Dec. 2024

Faculty of Mechanical Engineering, Delft University of Technology

Networking and matchmaking event for high-tech companies

ACADEMIC SERVICE

Member of editorial boards

- Adviser for Young Member, IFAC TC 1.5, Feb. 2025 - present
- Associate Editor, IEEE Conference on Decision and Control, 2026
- Associate Editor, IFAC World Congress 2026
- Associate Editor, European Control Conference 2026
- Associate Editor, IFAC NecSys 2025
- Member of Technical Committee, IEEE SoSE 2025

Organizer of events at international conferences

- 2 workshops at European Control Conference 2025 (**4+4 outstanding guest speakers**)
- 2 invited sessions at IEEE Conference on Decision and Control 2022 and 2024
- 1 workshop at IFAC World Congress 2023 (**6 outstanding guest speakers**)

Publicity co-chair at NextCom

Jan. 2025 – present

- **About:** NextCom supports career growth for young systems engineers and researchers

Founder and organizer of seminar series “DEITalks”

Oct. 2021- March 2023

Dept. Information Engineering, University of Padova

- **First initiative of this kind in the history of the department**
- **About:** Dissemination by and networking among PhD students and postdocs

Organizer of workshop “Control days”

Sept. 2022

Depts. Mathematics and Information Engineering, University of Padova

Founder and organizer of seminar series “Outing”

A. Y. 2020-2021

Dept. Information Engineering, University of Padova

PhD student's representative

AA. YY. 2019-2020, 2020-2021

Dept. Information Engineering, University of Padova

INVITED TALKS

Delft University of Technology (Graphs&Data@TUDelft seminar series)	Jan. 2025
Politecnico di Torino (IEEE CSS Italy Road2CDC seminar series)	Oct. 2024
KTH Royal Institute of Technology	April 2024
ETH Zurich	April 2023
MIT	Nov. 2022
Autonomy Talks (online seminar series)	Dec. 2021

PUBLICATIONS

- 24 co-authors from other institutions/countries
- 12/23 publications without PhD supervisor
- * Denotes equal contribution

Journal articles

1. L. Ballotta, G. Joseph, "Minimal actuator selection in linear time-invariant systems," *IEEE Trans. Autom. Control* 2026 [\[link\]](#)
2. L. Ballotta, N. Bastianello, R. Ferrari, K. H. Johansson, "Personalized and Resilient Distributed Learning Through Opinion Dynamics," *IEEE Trans. Control Netw. Syst.* 2025 [\[link\]](#)
3. L. Ballotta, J. Arbeláiz, V. Gupta, L. Schenato, M.R. Jovanović, "The Role of Communication Delays in the Control of Spatially Invariant Systems," *IEEE Trans. Autom. Control* 2025 [\[link\]](#)
4. B. Wolleswinkel, I. van Straalen, L. Ballotta, A. J. Gallo, R. Ferrari, "Periodic Sparse Control to Deter Undetectable Attacks on Over-Actuated Systems," *IEEE Control Syst. Lett.* 2025 [\[link\]](#)
5. O. E. Akgün, Á. Vékassy, L. Ballotta, M. Yemini, S. Gil, "Friedkin-Johnsen Is Distributed Gradient Descent," *IEEE Control Syst. Lett.* 2025 [\[link\]](#)
6. L. Ballotta, A. Peruffo, R. Ferrari, M. Mazo, "Fault detection via Output-based Barrier Functions," *European J. Control* 2025 [\[link\]](#)
7. L. Ballotta, R. Talak, "Safe Distributed Control of Multi-Robot Systems With Communication Delays," *IEEE Trans. Veh. Technol.* 2025 [\[link\]](#)
8. L. Ballotta, N. Dal Fabbro, G. Perin, L. Schenato, M. Rossi, G. Piro, "VREM-FL: Mobility-aware computation-scheduling co-design for vehicular federated learning," *IEEE Trans. Veh. Technol.* 2025 [\[link\]](#)
9. L. Ballotta, Á. Vékassy, S. Gil, M. Yemini, "Friedkin-Johnsen Model with Diminishing Competition," *IEEE Control Syst. Lett.* 2024 [\[link\]](#)
10. L. Ballotta, G. Joseph, I. R. Thete, "Pointwise-Sparse Actuator Scheduling for Linear Systems with Controllability Guarantee," *IEEE Control Syst. Lett.* 2024 [\[link\]](#)
11. L. Ballotta, G. Peserico, F. Zanini, P. Dini, "To compute or not to compute? Adaptive smart sensing in resource-constrained edge computing," *IEEE Trans. Netw. Sci. Eng.* 2024 [\[link\]](#)
12. L. Ballotta, G. Como, J.S. Shamma, L. Schenato, "Can competition outperform collaboration? The role of malicious agents," *IEEE Trans. Autom. Control* 2024 [\[link\]](#)
13. Y. Chang*, L. Ballotta*, L. Carlone, "D-Lite: Navigation-oriented compression of 3D Scene Graphs for Multi-Robot Collaboration," *IEEE Robot. Autom. Lett.* 2023 [\[link\]](#)
14. L. Ballotta, V. Gupta, "Faster consensus via a sparser controller," *IEEE Control Syst. Lett.* 2023 [\[link\]](#)
15. L. Ballotta, M.R. Jovanović, L. Schenato, "Can decentralized control outperform centralized? The role of communication latency," *IEEE Trans. Control Netw. Syst.* 2023 [\[link\]](#)
16. E. Rossi, M. Tognon, L. Ballotta, R. Carli, J. Cortés, A. Franchi, L. Schenato, "Coordinated multi-robot trajectory tracking control over sampled communication," *Automatica* 2023 [\[link\]](#)

17. L. Ballotta, L. Schenato, L. Carlone, "Computation-communication trade-offs and sensor selection in real-time estimation for processing networks," *IEEE Trans. Netw. Sci. Eng.* 2020 [\[link\]](#)

Conference papers

1. J. Liu, Y. Yang, L. Ballotta, F. Li, Y. Tang, R. Carli, "Submodular Multi-Agent Policy Learning for Online Distributed Task Allocation in Open Multi-Agent Systems," *NeurIPS* 2026 [\[link\]](#)
2. Y. Wang, L. Ballotta, X. Cao, R. Carli, L. Schenato, "Pursuing Optimal Stepsize in Adaptive Gradient-Based Quadratic Optimization," *IFAC World Congress* 2026 [\[link\]](#)
3. L. Ballotta, J. Arbelaiz, V. Gupta, L. Schenato, M. R. Jovanović, "Control of Reaction-Diffusion Processes under Communication Delays," *European Control Conf.* 2024 [\[link\]](#)
4. L. Ballotta, M. Yemini, "The role of confidence for trust-based resilient consensus," *American Control Conf.* 2024 [\[link\]](#)
5. L. Ballotta, G. Como, J.S. Shamma, L. Schenato, "Competition-based resilience in distributed quadratic optimization," *IEEE Conf. Decis. Control* 2022 [\[link\]](#)
6. L. Ballotta, G. Peserico, F. Zanini, "A reinforcement learning approach to sensing design in resource-constrained wireless networked control systems," *IEEE Conf. Decis. Control* 2022 [\[link\]](#)
7. V. Tripathi*, L. Ballotta*, L. Carlone, E. Modiano, "Computation and communication co-design for realtime monitoring and control in multi-agent systems," *WiOpt* 2021 [\[link\]](#)
8. L. Ballotta, M.R. Jovanović, L. Schenato, "Optimal network topology of multi-agent systems subject to computation and communication latency," *MED* 2021 [\[link\]](#)
9. L. Ballotta, L. Schenato, L. Carlone, "From sensor to processing networks: Optimal estimation with computation and communication latency," *IFAC World Congress* 2020 [\[link\]](#)

SKILLS

- English: C1 (IELTS score band 8, issued November 2024)
- Problem solving, critical thinking
- Flexible, adaptive, open-minded
- Leadership, collaboration, teamwork
- Dissemination, event organization
- Time and deadline management

OTHER WORK EXPERIENCES

Mentor for EduTutor (collaborator as external expert) Feb. – Dec. 2025

- I used to advise and support Italian students who apply to Dutch universities

Website developer Jan. – June 2023

- I updated the new website of the control group of the University of Padova

Volleyball trainer June 2012, June 2011

- I was a short-term trainer at a sport summer camp for kids (age 9 – 11)

VOLOUNTEERING

VP of communications at FAIL! Inspiring Resilience June 2020 – Oct. 2021
Cambridge, MA USA

- **About:** FAIL! aims to destigmatize failure and build resilience in academia and beyond
- **Responsibilities:** Manager of social media, newsletter, online event advertising

VP of communications at MIT Visiting Student Association

March – Sept. 2020

Cambridge, MA USA

- **About:** MIT VISTA organizes events and activities for international visiting students
- **Responsibilities:** Manager of newsletter for event advertising, co-organizer of events

Member of local parish Christian association (Azione Cattolica)

2011 – 2017

Ponte San Nicolò, Italy

- VP (elect) and responsible person of activities for adolescents (age 14 – 17)
- Organization of summer camps, management of logistics and finances (10k – 15k EUR)